

Parallel Scenario

What is the average water consumption in liters of the top 3 S&P 500 companies in 2023, considering only UK branches?

Extract one value from each non-relational table, then aggregate

Company X		
2023 indicators per employee	Branch location	
	UK	US
Weekly water usage 2022	5.89 gallons	7.16 gallons
Weekly water usage 2023	5.60 gallons	--
Perturbations: Pivot + unit change + n.a.		

Company Y		
Branch locations 2023	Weekly liters (per employee)	
	Water	Oil
France	22	31
UK	21.5	not available
Perturbations: Pivot + unit in header + n.a.		

Company Z			
Weekly usage per employee		Rbnach citise	
		Lodnon (UK)	Tokoy (Jaapn)
Waetr	2022	0.022 m^3	0.031 m^3
	2023	0.024 m^3	0.028 m^3
Perturbations: Row/col pivot + unit change + OCR			

Conversion: 5.6 gallons $\rightarrow$ 21.2 liters, 0.024 $m^3 \rightarrow$ 24 liters $\rightarrow$ Answer: 22.23 liters
avg(21.2, 21.5, 24)

Sequential Scenario

*What is the average 2023 investment of the company where John Doe is CEO?
Restrict the calculation to investments done in AI and the financial sector*

Follow implicit lookup keys across perturbed tables

CEO names		
Year	Compaines	
	X	Y
	2022	John Doe
2023	Jnae Mya	n.a.
Perturbations: Pivot + n.a. + OCR		

Facility assignment			
Facilities	Compaines		
	X	Y	
	Primary	Atlanta Lab	Seoul Lab
Backup	n.a.	Busan Hub	
Perturbations: Pivot + n.a. + empty column			

R&D expenses			
Costs (2023)		Investments (millions)	
		AI Research	Finance Research
Net cost	Atlanta Lab	42.1	38.9
	Seoul Lab	53.8 - as written in [1]	n.a.
Perturbations: Row/Col pivot + text insertion			

propagate values
Company X $\rightarrow$ Atlanta Lab $\rightarrow$ avg(42.1, 38.9) $\rightarrow$ Answer: 40.5